

Functional equations

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1 Classical introduction

In a functional equation, one tries to determine all functions satisfying a certain equation. One may know some functions like trigonometric functions, polynomials, exponential functions etc., but one cannot list all functions as such. The definition is very general.

Definition 1.1. A function $f: A \rightarrow B: a \mapsto f(a)$ is any mapping that assigns to every element $a \in A$ an element $f(a) = b \in B$.

For finite sets, there are $|B|^{|A|}$ many such functions. In that case, one could list all the functions by hand.

Nevertheless, this may soon be infeasible.

Example 1.2. Determine all $f: \frac{\mathbb{Z}}{5\mathbb{Z}} \rightarrow \frac{\mathbb{Z}}{5\mathbb{Z}}: x \mapsto f(x)$ satisfying $f(a+b) = f(a)^2 + f(b^2)$ for every $a, b \in \frac{\mathbb{Z}}{5\mathbb{Z}}$.

There are 5^5 many possibilities to verify if checking all of them.

Nevertheless, this one is easily solved by using some basic properties and choices of (a, b) .

- Determine $f(0)$.
- Restrict the possibilities of outcomes for $f(a)$.
- Conclude.

In particular, we notice that one given equation can reduce the set of functions satisfying it a lot. When we know the function has also an additional property or belongs to a specific class of functions, there are even more tools and elementary properties to use. E.g.

Observation 1.3. If P is a polynomial with integer coefficients, i.e., $P(X) \in \mathbb{Z}[X]$, then $a - b \mid P(a) - P(b)$ for every $a, b \in \mathbb{Z}$.

Theorem 1.4. Every polynomial can be linearly over $\mathbb{C}$. Every polynomial in $\mathbb{R}[X]$ is the product of degree one and degree two polynomials over $\mathbb{R}$.

Theorem 1.5. (Rolle) If $f: [a, b] \rightarrow \mathbb{R}$ (where $b > a$) is a differentiable function satisfying $f(a) = f(b)$, there exists $a < c < b$ for which $f'(c) = 0$.

2 Summary basic ideas

- Substitutions
- Induction
- Prove claims about injectivity/ surjectivity (and thus bijectivity)
- Finding zeros or fixed points
- Use properties if given; continuity/ polynomials
- Solve for a dense subset of the domain
- Recurrent relations; if $f(f(n)) = af(n) + b$, this can be repeated to express $f^{(k)}(n)$ as a linear combination of $f(n)$ and 1.
- Reduce to known equations; Cauchy's equation:

$$f(x+y) = f(x) + f(y) \quad \forall x, y \in S$$

If $S = \mathbb{Q}$, $f(x) = cx$.

If $S = \mathbb{Q}[\sqrt{2}]$, $f(a + b\sqrt{2}) = ac + bd$ ($a, b \in \mathbb{Q}$, $c, d \in \mathbb{Q}[\sqrt{2}]$).

If $S = \mathbb{R}$, many solutions, determined by base of $\mathbb{R}$ over $\mathbb{Q}$. With e.g. continuity or monotonicity, we know $f(x) = cx$.

- Write f as the sum of an even - and an odd function
- Difference with expected solution (if unique); if f_0 is the unique solution, substitute $f(x) = f_0(x) + g(x)$ and prove that g is the zero function.
- W.l.o.g.'s by invariant under operation; if $f(x) + c$ or $cf(x)$ are solutions whenever f is, we can choose the value of $f(0)$ or $f(1)$

Warning: Determining all functions, implies you prove **both** certain functions satisfy the properties, and other functions do not. So also check that the final subset of functions you obtained all satisfy the question.

3 Intro on functional equations related to number theory

Functional equations with a number theoretic flavour usually combine ideas from the classical FE-world with some theorems from number theory. They can depend on simple division, or on the rich structure of primes.

As is often the case, there is no method that solves all problems at once. But for a functional equation of the form $LHS \mid RHS$, we can list some ideas.

- Find expression like $f(a) \mid LHS \mid nf(a)$ where n has not too many prime factors.
- Prove that infinitely many primes, or no prime divide a fixed expression like $f(n) - n$ or $f(n+1) - f(n)$.
- Determine f on primes or prime powers
- Number theoretic theorems; e.g. if $f(a) \pmod{p}$ is a quadratic residue for all primes p , then $f(a)$ is a perfect square.

4 Exercises

✓ warm-up 1 Find all $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x+y) = f(x) + y$ for every $x, y \in \mathbb{R}$.

✓ NT There exists a polynomial $P(X) \in \mathbb{Z}[X]$ such that $P(0) = 1$, $P(2) = 3$ and $P(4) = 9$?

✓ Romania '86 Let $f, g: \mathbb{N} \rightarrow \mathbb{N}$ with f surjective and g injective, and $f(n) \geq g(n)$ for every $n \in \mathbb{N}$. Prove that $f(n) = g(n)$ for every $n \in \mathbb{N}$.

✓ IMC '99 There exists a bijection $f: \mathbb{N} \rightarrow \mathbb{N}: n \mapsto f(n)$ for which $\sum_{n=1}^{\infty} \frac{f(n)}{n^2}$ is finite?

✓ IMC '99 Find all strictly monotonic functions $f: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$(f(x))^2 \geq f(x+y)(f(x)+y) \quad \forall x, y > 0.$$

✓ Dutch TST '20 Determine all polynomials $P(x)$ with real coefficients that apply $P(x^2) + 2P(x) = P(x)^2 + 2$.

✓ FMO '22 Find all $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$f(m+n) + f(m)f(n) = n^2 f(m) + m^2 f(n) + (m+n)^2 - m^2 n^2, \quad \forall m, n \in \mathbb{Z}.$$

✓ Euler '25 Find all functions $f: \mathbb{Q}[\sqrt{2}] \rightarrow \mathbb{Q}[\sqrt{2}]$ for which $f(x+y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{Q}[\sqrt{2}]$.

IMSCSL '24 Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ for which the following conditions hold for every $n \in \mathbb{N}$:

- $f(n+1) \mid n$,
- $f\left(f\left(\frac{n}{f(n+1)}\right) + f(n+1)\right) = n$.

Here $\mathbb{N}$ are the positive integers.

ELMO SL '14 Let $\mathbb{N}$ denote the set of positive integers. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that:

- The greatest common divisor of the sequence $f(1), f(2), \dots$ is 1.
- For all sufficiently large integers n , we have $f(n) \neq 1$ and

$$f(a)^n \mid f(a+b)^{a^{n-1}} - f(b)^{a^{n-1}}$$

for all positive integers a and b .

5 Hints

- Can you guess the functions satisfying it? Can you prove it has roughly this form?
- Do you know some structure of P by knowing $P(0)$?
- Assume the contrary. Consider the set $A = \{n : f(n) \neq g(n)\}$.
- What do you know about the values $f(x)$ for $n+1 \leq x \leq 3n$? Alternative: What do you know about $\sum_{n=1}^k \frac{f(n)}{n^2}$?
- Try to prove that $f(x)$ is sufficiently fast decreasing to obtain a contradiction.
- Let $Q(x) = P(x) - 1$.
- Note that $f(n) = f(n) - n^2$ satisfies

$$g(m+n) + g(m)g(n) = 0, \quad \forall m, n.$$

- It is sufficient to know $f(1)$ and $f(\sqrt{2})$.

IMSSSL '24 First determine $f(n)$ for $n \geq 2$

ELMO SL '14 Start proving that $f(a) \mid a$ for every $a \in \mathbb{N}$.

6 Solutions and solution sketches

Problem 6.1 (warm-up). Find all $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x+y) = f(x) + y$ for every $x, y \in \mathbb{R}$.

Proof. Choose $x = 0$. Then $f(y) = y + f(0)$. Now since every function $f(x) = x + c$ works, these are exactly the solutions. $\square$

Problem 6.2 (NT). There exists a polynomial $P(X) \in \mathbb{Z}[X]$ such that $P(0) = 1, P(2) = 3$ and $P(4) = 9$?

Proof. No. Note that $P(X) = XQ(X) + 1$ for some $Q \in \mathbb{Z}[X]$. Here $Q(2) = 1$ and $Q(4) = 2$. So $4 - 2 \nmid Q(4) - Q(2)$, from which we conclude. $\square$

Problem 6.3 (Romania '86). Let $f, g : \mathbb{N} \rightarrow \mathbb{N}$ with f surjective and g injective, and $f(n) \geq g(n)$ for every $n \in \mathbb{N}$. Prove that $f(n) = g(n)$ for every $n \in \mathbb{N}$.

Proof. Assume not. Consider the set $A = \{n : f(n) \neq g(n)\}$. Let $B = \{f(n) \mid n \in A\}$ and $C = \{g(n) \mid n \in A\}$. Then by definition, $\min B > \min C$ on the one hand, but $B = C$ on the other hand. $\square$

Problem 6.4 (IMC '99). There exists a bijection $f : \mathbb{N} \rightarrow \mathbb{N} : n \mapsto f(n)$ for which $\sum_{n=1}^{\infty} \frac{f(n)}{n^2}$ is finite?

Proof. Decrease values above k if necessary, and conclude that $\sum_{n=1}^k \frac{f(n)}{n^2} \geq \sum_{n=1}^k \frac{1}{n}$. The latter goes to infinity as $k \rightarrow \infty$, from which the conclusion follows. $\square$

Problem 6.5 (IMC '99). Find all strictly monotonic functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$(f(x))^2 \geq f(x+y)(f(x)+y) \quad \forall x, y > 0.$$

Proof. Note that f is decreasing and that the inequality is equivalent with $f(x+y) \leq f(x) - \frac{f(x)y}{f(x)+y}$. Choose $y = \frac{1}{n} < f(x+1)$. Then $f(x+ky) < f(x+(k-1)y) - \frac{y}{2}$ for every $1 \leq k \leq n-1$ and thus $f(x+1) \leq f(x) - 1/2$. Hence $f(x+1+2n) < 0$, contradiction. $\square$

Problem 6.6 (Dutch TST '20). Determine all polynomials $P(x)$ with real coefficients that apply $P(x^2) + 2P(x) = P(x)^2 + 2$.

Proof. Let $Q(x) = P(x) - 1$, then $Q(x^2) = Q(x)^2$. If Q is not a constant function and has a (possibly complex) root r different from 0 and 1, then so is $\sqrt{r}$, leading to infinitely many roots. If $Q(1) = 0$, also $Q(-1) = 0$ and thus a contradiction is obtained. Hence $Q(x) = cx^n$, and $c \in \{0, 1\}$ (since $c^2 = c$), and these also work. We conclude that $P(X) = 1$ or $P(X) = x^n + 1$ for some $n \geq 0$. $\square$

Problem 6.7 (FMO '22). Find all $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$f(m+n) + f(m)f(n) = n^2 f(m) + m^2 f(n) + (m+n)^2 - m^2 n^2, \quad \forall m, n \in \mathbb{Z}.$$

Proof. The function $g(n) = f(n) - n^2$ satisfies $g(m+n) + g(m)g(n) = 0$, $\forall m, n$. In particular $g(n)(1+g(0)) = 0$. If $g(0) \neq -1$, then $g \equiv 0$.

If $g(0) = -1$, choosing $m = -n$ results in $g(m) = g(n) = \pm 1$. Here $m = n$ implies that $g(2n) = -1$. Finally, we have $g(2n+1) = g(1) \in \{-1, 1\}$.

The solutions are $f(n) = n^2$, $f(n) = n^2 - 1$ and $f(n) = n^2 - (-1)^n$. Finally, one checks those three functions. $\square$

Problem 6.8 (Euler '25). Find all functions $f: \mathbb{Q}[\sqrt{2}] \rightarrow \mathbb{Q}[\sqrt{2}]$ for which $f(x+y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{Q}[\sqrt{2}]$.

Proof. We know by Cauchy's equation (classical form) that $f(a + b\sqrt{2}) = af(1) + bf(\sqrt{2})$. Now $f(2) = f(\sqrt{2})^2$ and $2f(2) = f(4) = f(2)^2$. This implies that $f(2) \in \{0, 2\}$. Furthermore, in the second case $f(\sqrt{2}) = \pm\sqrt{2}$.

We conclude that the functional equation has three solutions; $f(z) = 0$, $f(z) = z$ and $f(z) = \bar{z}$ for all $z \in \mathbb{Q}[\sqrt{2}]$. Here $a + b\sqrt{2} = a - b\sqrt{2}$. The first two trivially satisfy the equalities. The third one is also checked by direct verification. $\square$

Problem 6.9 (IMSCSL '24). Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ for which the following conditions hold for every $n \in \mathbb{N}$:

- $f(n+1) \mid n$,
- $f\left(f\left(\frac{n}{f(n+1)}\right) + f(n+1)\right) = n$.

Proof. Notice $f(2) = 1$. By induction, we will prove that $f(n+1) = n$ for every $n \geq 1$. Assuming this is true for $1 \leq k \leq n-1$, if $f(n+1) \neq n$, then $f\left(\frac{n}{f(n+1)} - 1 + f(n+1)\right) = n$ and thus $\frac{n}{f(n+1)} + f(n+1) > n+1$, so $f(n+1) > n$. This is a contradiction with $f(n+1) \mid n$. Finally, $f(1)$ is determined by the second part for large n and the previous conclusion: $f(1) = 1$. Thus $f(x) = x - 1 + 1_{x=1}$, and this function satisfies the equation. $\square$

Problem 6.10 (ELMO SL '14). Let $\mathbb{N}$ denote the set of positive integers. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that:

- The greatest common divisor of the sequence $f(1), f(2), \dots$ is 1.
- For all sufficiently large integers n , we have $f(n) \neq 1$ and

$$f(a)^n \mid f(a+b)^{a^{n-1}} - f(b)^{a^{n-1}}$$

for all positive integers a and b .

Proof. Notice that the smallest prime factor of $f(1)$ divides all others (by second property with $a = 1$ by induction) and thus $f(1) = 1$.

Next, fix a . Note that $p \nmid f(a+1)$ for every prime divisor p of a , by choosing $b = 1$ in the second equation.

Let d be the order of $f(a+1) \bmod p$. By LTE, if p is odd, then

$$nv_p(f(a)) \leq v_p(f(a+1)^d - 1) + (n-1)v_p(a) \leq (p-1)\log f(a+1) + (n-1)v_p(a).$$

If $p = 2$, then

$$nv_2(f(a)) \leq v_2(f(a+1) - 1) + v_2(f(a+1) + 1) - 1 + (n-1)v_2(a).$$

If $v_p(f(a)) > v_p(a)$, for n sufficiently large this leads to a contradiction.

We conclude that $f(a) \mid a$ for all $a \in \mathbb{N}$.

In particular, for all sufficiently large primes p , $f(p) = p$.

Now

$$p^n = f(p)^n \mid f(b+p)^{p^{n-1}} - f(b)^{p^{n-1}},$$

for n sufficiently large.

Fix b . When $\gcd(b, p) = 1$ and thus $p \nmid f(b)$ (being a divisor of b), the order of $\frac{f(b+p)}{f(b)}$ equals $\gcd(p-1, p^{n-1}) = 1$, i.e. $p \mid f(b+p) - f(b)$.

This inductively implies $p \mid f(b+mp) - f(b)$. By Dirichlet, there are (infinitely many) primes of the form $b+mp$, and thus $p \mid b - f(b)$. Hence $f(b) = b$.

The identity function indeed satisfies the divisibility condition (again check the prime divisors $p \mid a$ with LTE). $\square$